

Architectural Vision

AI Disclosure: Generative AI was used when writing this document.

The chat logs can be viewed here: <https://share.gemini.google/xuJEuZ0Lf81t>

Business Context

SolarSphere Energy is a multinational renewable energy producer that operates utility-scale solar energy facilities across Europe, the Middle East, Africa (EMEA), North America, and the Asia-Pacific (APAC) regions. The company's portfolio includes utility-scale photovoltaic (PV) farms composed of high-capacity (500-600 watt) solar panels, distributed solar generation facilities, and utility-scale Battery Energy Storage Systems (BESS) deployed to manage grid stabilisation and energy dispatch.

To support global energy delivery, SolarSphere owns and maintains critical grid interconnection infrastructure including substations, transformers, high-voltage transmission lines, and cross-border power cables. Regional Operations Control Centers (ROCCs) monitor and coordinate power generation and transmission in real-time using Supervisory Control and Data Acquisition (SCADA) systems, and on-site telemetry sensors and actuators. They also own fleets of vehicles for field service operations such as maintenance, inspections, and emergency repairs across all their sites.

SolarSphere generates revenue by supplying electricity to utility companies, commercial clients, through Power Purchase Agreements (PPA), as well as participating directly in regional energy markets. Managing this globally distributed system has led to many technical challenges. Developing a unified enterprise architecture will help to overcome issues such as optimising energy generation and weather forecasting, streaming, storing, and analysing large quantities of telemetry data, and meeting regional regulatory and Environmental, Social, and Governance (ESG) compliance standards.

Business Drivers

The design of SolarSphere Energy's target architecture is guided by key business goals that focus on improving plant efficiency, asset management, and regulatory compliance.

Firstly, the company wishes to maximise energy generation by improving the Performance Ratio (PR) of their solar farms and distributed solar generation facilities. PR is a metric which compares the actual energy produced at a facility to the theoretical maximum energy produced under ideal conditions. Optimising this metric requires real-time asset monitoring combined with weather and energy generation forecasting using data analytics.

Secondly, increasing asset availability is critical to reducing unplanned downtime and lowering operational expenses in the long term. By combining real-time asset monitoring, asset lifecycle management, and predictive analytics, the company can change from reactive asset maintenance towards proactive maintenance scheduling, which will directly decrease maintenance costs.

Thirdly, the company aims to optimise the utilisation of their Battery Energy Storage Systems. Monitoring from Regional Operations Control Centres, combined with smart grid technologies, enables automated energy dispatch in response to grid demand and regional market fluctuations.

Finally, data flow from the company's OT/IT systems needs to be securely integrated to enable real-time visibility, automated workflows and cross-functional operations. Connecting edge sensors, local and control centre SCADA systems, and Battery Management Systems (BMS) with enterprise IT systems such as Enterprise Resource Planning (ERP) and Enterprise Asset Management (EAM) systems would allow for operational automation and aid in generating accurate sustainability and Environmental, Social, and Governance (ESG) reporting.

Architectural Principles

To ensure the architecture aligns with the company's business objectives and technical requirements, all architectural choices will be governed by five core principles.

Renewable Operations First – Due to the critical nature of the business as an energy provider, the operational continuity of power generation and battery storage facilities takes precedence over administrative IT and corporate reporting. In the event of network disruptions between facilities and Regional Operations Control Centres, local SCADA and Battery Management Systems (BMS) must be able to maintain uninterrupted, autonomous local operational control. Sensor telemetry data should be stored locally and automatically synced with the central data lake.

Secure OT/IT Integration – There should be a strict separation between operational plant networks and corporate IT environments. Data flow between these zones must pass through a dedicated Industrial DMZ (IDMZ) and secure gateway services to prevent external cyber security threats affecting physical equipment.

Real-Time Visibility – Live status telemetry of all connected assets should be available to control centre operators with minimal latency. High-speed data stream processing is necessary to enable immediate operational responses to power grid fluctuations.

Data-Driven Decision Making – Critical business actions, such as energy trading, asset maintenance scheduling, and weather and generation forecasting, should be informed by real-time asset telemetry processed on cloud platforms. To support this, the architecture relies on centralised cloud analytics platforms and unified data definitions across all operational facilities.

Sustainability by Design – Environmental and carbon metrics are automatically captured directly from sensors on operational assets. Automatically tracking clean energy generation and emissions data at the asset level streamlines regulatory compliance and eliminates the manual effort of generating ESG reports for stakeholders.

Scope

The scope of this project covers the Information Architecture and the Application Architecture necessary to support the company's core operations. This includes solar energy generation, Battery Energy Storage Systems (BESS), grid interconnection, real-time telemetry processing, cloud data analytics, weather and energy generation forecasting, asset availability and predictive maintenance, and automated sustainability and ESG reporting. Physical plant engineering, civil/electrical construction details, and hardware vendor selection are outside the scope of this project.

Stakeholders

The enterprise architecture involves both internal and external stakeholders involved in the company's operations, governance, and commercial activities. External stakeholders include utility companies, commercial and Power Purchase Agreement (PPA) clients, regional energy market operators, regulatory bodies, and technology suppliers providing solar panels, battery systems, grid interconnection hardware, and enterprise software platforms. Internal stakeholders include the company's board of directors, shareholders, control centre operators, facility and plant engineers, field service maintenance technicians, and commercial enterprise employees.

Business Capability Model

The business capability model for SolarSphere Energy is structured into three primary capability domains.

The first domain, **core operational capabilities**, is comprised of the activities required to generate, store and manage energy assets. This domain includes:

- *Energy Generation Management* which covers real-time plant monitoring, performance ratio optimisation, curtailment, and outage handling.
- *Data Analysis and Forecasting* which covers weather, irradiance, energy generation, and market forecasts.
- *Battery Storage Optimisation* which covers state-of-charge monitoring, charge and discharge scheduling, and cell degradation management.
- *Grid Integration* which covers automated energy dispatch, grid-code compliance, and grid interconnection control.
- *Asset Management* which covers physical asset inspections, work order execution, and predictive maintenance.
- *Industrial Cybersecurity and Resilience* which covers OT/IT integration and secure telemetry communication.
- *Real-Time Telemetry Processing* which covers communication, storage and processing of real-time plant sensor data.

The second domain, **strategic capabilities**, is comprised of the activities which aid in commercial growth, market participation, and corporate governance. This domain includes:

- *Power Purchase Agreements (PPA) management.*
- *Commercial contract management.*
- *Regional energy market participation.*
- *Sustainability and ESG reporting.*

The third domain, **supporting capabilities**, include the foundational infrastructure and administrative functionality necessary for global operations. This domain includes:

- Enterprise Management, which covers Enterprise Resource Planning (ERP) functions such as supply chain management, asset procurement, and vendor management.

High-Level Roadmap

The enterprise architecture involves three phases over an 18-month period, each period lasting six months.

The first phase, **Foundation and Standardisation**, establishes IDMZs and secure gateway services, high speed WANs, and master data definitions for assets and sites.

The second phase, **Centralisation and Cloud Analytics**, deploys the central cloud data lake, establishes real-time telemetry streaming from plants and BESS units, and implements basic forecasting models using cloud infrastructure.

The third phase, **Operational Optimisation**, deploys predictive asset maintenance, automated BESS energy dispatch, and real-time automated sustainability and ESG reporting.